

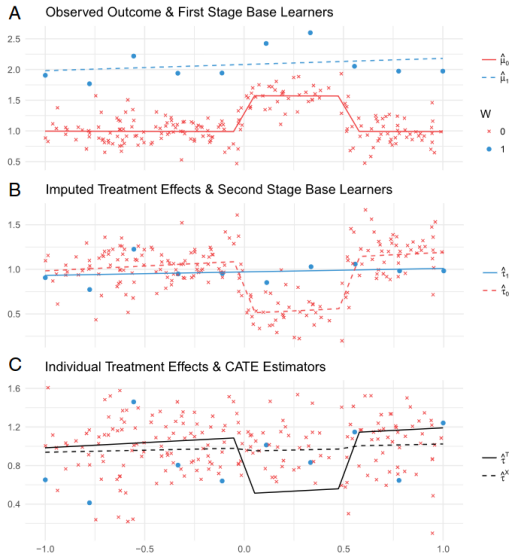
# Causal Inference and ML:

## Robust Estimation of Treatment Heterogeneity: CATE, Meta-Learners, Confounding-Robust Methods and Causal Forests

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# Visualizing Outcomes, Learners, CATE



## Quiz 1A: Technical Identification

Under unconfoundedness, which formula identifies the CATE?

1.  $\mathbb{E}[Y_i \mid X_i = x, W_i = 1] - \mathbb{E}[Y_i \mid X_i = x, W_i = 0]$
2.  $\mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]$
3.  $\sum_{i=1}^n (Y_i - \bar{Y})(W_i - \bar{W})$
4.  $\mathbb{E}[Y_i(W_i) \mid W_i = 1]$

## Quiz 1A Solution

1.  $\mathbb{E}[Y_i \mid X_i = x, W_i = 1] - \mathbb{E}[Y_i \mid X_i = x, W_i = 0]$
2.  $\mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]$  — this is the ATE, not CATE
3.  $\sum_{i=1}^n (Y_i - \bar{Y})(W_i - \bar{W})$
4.  $\mathbb{E}[Y_i(W_i) \mid W_i = 1]$

## Quiz 1B: Vocabulary

Which term describes the bias when an ML model regularizes treatment and control functions differently due to unequal sample sizes?

1. Selection Bias
2. Regularization Bias
3. Omitted Variable Bias
4. SUTVA Violation - Stable Unit Treatment Value Assumption

# Quiz 1B Solution

1. Selection Bias
2. Regularization Bias
3. Omitted Variable Bias
4. SUTVA Violation - Stable Unit Treatment Value Assumption

# The Goal: Conditional Average Treatment Effect (CATE)

- ▶ **Setup:** Observed data  $(X_i, W_i, Y_i)$ , where  $X_i \in \mathbb{R}^d$  are features,  $W_i \in \{0, 1\}$  is treatment,  $Y_i \in \mathbb{R}$  is outcome. Potential outcomes  $Y_i(0), Y_i(1)$  are unobserved.
- ▶ **CATE:**

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$$

This is the treatment effect for individuals with features  $x$  — it varies with  $x$ .

- ▶ **Identification under Unconfoundedness:**

- ▶ Assumption:  $\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i$  (no unmeasured confounding)
- ▶ Then:

$$\tau(x) = \underbrace{\mathbb{E}[Y_i \mid X_i = x, W_i = 1]}_{\mu^{(1)}(x)} - \underbrace{\mathbb{E}[Y_i \mid X_i = x, W_i = 0]}_{\mu^{(0)}(x)}$$

- ▶ CATE is identified as the conditional mean difference between treated and control.

# Two Key Obstacles: Regularization and Confounding

## ► Regularization Bias:

- ML models optimize for prediction, not causal effects.
- With  $n_1 \neq n_0$ , regularization affects treatment and control models differently, leading to biased CATE estimates:

$$\hat{\tau}(x) = \hat{\mu}^{(1)}(x) - \hat{\mu}^{(0)}(x) \neq \tau(x)$$

- Can produce "hallucinated" heterogeneity even when none exists.

## ► Confounding Bias:

- In observational studies,  $W$  correlates with  $X$ , so the naive difference in means is biased:

$$\mathbb{E}[Y|W=1] - \mathbb{E}[Y|W=0] \neq \tau$$

- The bias is selection bias from non-random treatment assignment — units differ systematically before treatment.



## Metalearners for estimating heterogeneous treatment effects using machine learning

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Contributed by Bin Yu, December 18, 2018 (sent for review March 16, 2018; reviewed by Jake Bowers and Dylan Small)

There is growing interest in estimating and analyzing heterogeneous treatment effects in experimental and observational studies. We describe a number of metaalgorithms that can take

problems that can be solved with any regression or supervised learning method.

The most common metaalgorithm for estimating heteroge-

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### GENERALIZED RANDOM FORESTS

BY SUSAN ATHEY\*, JULIE TIBSHIRANI<sup>†</sup> AND STEFAN WAGER\*

*Stanford University\* and Elasticsearch BV<sup>†</sup>*

We propose generalized random forests, a method for nonparametric statistical estimation based on random forests (*Biometrika* **106**, 2019).

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## GENERALIZED RANDOM FORESTS

Susan Athey, Julie Tibshirani and Stefan Wager

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## Quiz 2A: Learner Types

Which meta-learner fits a single model  $\mu(x, w)$  and estimates  $\tau(x)$  by querying it at  $w = 1$  and  $w = 0$ ?

1. T-Learner
2. X-Learner
3. S-Learner
4. R-Learner

## Quiz 2A Solution

1. T-Learner — fits two separate models
2. X-Learner — uses imputation and propensity weighting
3. S-Learner — single model
4. R-Learner

## Quiz 2B: X-Learner Imputation

What are the pseudo-effects  $D_i^{(1)}$  used to train  $\hat{\tau}^{(1)}(x)$  in the X-learner?

1.  $Y_i - \hat{\mu}^{(1)}(X_i)$
2.  $Y_i - \hat{\mu}^{(0)}(X_i)$
3.  $W_i - \hat{e}(X_i)$
4.  $\hat{\mu}^{(1)}(X_i) - \hat{\mu}^{(0)}(X_i)$

## Quiz 2B Solution

1.  $Y_i - \hat{\mu}^{(1)}(X_i)$
2.  $Y_i - \hat{\mu}^{(0)}(X_i)$  — treated outcome minus imputed counterfactual control
3.  $W_i - \hat{e}(X_i)$
4.  $\hat{\mu}^{(1)}(X_i) - \hat{\mu}^{(0)}(X_i)$

# S-Learner and T-Learner

- ▶ **S-Learner:** Treats  $W$  as a feature in a single model
$$\mu(x, w) = \mathbb{E}[Y|X = x, W = w]$$
  - ▶ Train one model on all data; predict at  $W = 1$  and  $W = 0$
  - ▶  $\hat{\tau}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$
  - ▶ Risk: regularization may shrink the effect of  $W$ , biasing  $\hat{\tau}$
- ▶ **T-Learner:** Trains two independent models on each group
  - ▶  $\hat{\mu}^{(0)}(x)$ : fit on control units;  $\hat{\mu}^{(1)}(x)$ : fit on treated units
  - ▶  $\hat{\tau}(x) = \hat{\mu}^{(1)}(x) - \hat{\mu}^{(0)}(x)$
  - ▶ Risk: different regularization per group can hallucinate heterogeneity
- ▶ **Key difference:** S-learner pools all data; T-learner separates groups. Both estimate  $\tau(x)$  by differencing predictions.

# The X-Learner Strategy

- ▶ Proposed by Künzel et al. (2019) to reduce regularization bias.
- ▶ **Step 1:** Estimate response functions (T-learner style):  $\hat{\mu}^{(0)}(x)$  on control units,  $\hat{\mu}^{(1)}(x)$  on treated units.
- ▶ **Step 2:** Impute pseudo-effects for each unit:
  - ▶ Treated:  $D_i^{(1)} = Y_i - \hat{\mu}^{(0)}(X_i)$  (actual outcome minus imputed control)
  - ▶ Control:  $D_i^{(0)} = \hat{\mu}^{(1)}(X_i) - Y_i$  (imputed treated minus actual outcome)
- ▶ **Step 3:** Fit effect models on pseudo-effects:
  - ▶  $\hat{\tau}^{(1)}(x)$ : regress  $D_i^{(1)}$  on  $X_i$  for treated units
  - ▶  $\hat{\tau}^{(0)}(x)$ : regress  $D_i^{(0)}$  on  $X_i$  for control units
- ▶ **Final estimate** (weighted by propensity score):

$$\hat{\tau}(x) = \hat{e}(x)\hat{\tau}^{(1)}(x) + (1 - \hat{e}(x))\hat{\tau}^{(0)}(x)$$

The propensity score  $\hat{e}(x)$  upweights the more reliable estimate in each region.

# Comparison of Meta-Learners

|                 | <b>S-Learner</b>                        | <b>T-Learner</b>                              | <b>X-Learner</b>                                                      |
|-----------------|-----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------|
| <b>Idea</b>     | Single model with $W$ as feature        | Two separate models                           | Imputed pseudo-effects, propensity weighting                          |
| <b>Formula</b>  | $\hat{\tau}(x) = \mu(x, 1) - \mu(x, 0)$ | $\hat{\tau}(x) = \mu^{(1)}(x) - \mu^{(0)}(x)$ | $\hat{e}(x)\hat{\tau}^{(1)}(x) + (1 - \hat{e}(x))\hat{\tau}^{(0)}(x)$ |
| <b>Pros</b>     | Simple; uses all data                   | Flexible; handles different baselines         | Reduces regularization bias                                           |
| <b>Cons</b>     | Regularization bias; may over-smooth    | Can hallucinate heterogeneity                 | More complex; needs good $\hat{e}(x)$                                 |
| <b>Best for</b> | Small effects, simple settings          | Large samples, different baseline functions   | Imbalanced treatment/control sizes                                    |



# Route towards Causal Forests

## Quiz 3A: Semiparametric Vocabulary

In Robinson's transformation,  $\mu^{(0)}(x)$  is called a:

1. Target Parameter
2. Treatment Effect
3. Propensity Score
4. Nuisance Component

## Quiz 3A Solution

1. Target Parameter
2. Treatment Effect
3. Propensity Score
4. Nuisance Component — must be estimated but is not the quantity of interest

# Robinson's Transformation

- ▶ **Partial Linear Model:** Assumes constant treatment effect  $\tau$ :

$$\mathbb{E}[Y \mid X = x, W = w] = \mu^{(0)}(x) + w\tau$$

where  $\mu^{(0)}(x)$  is a flexible nuisance baseline function.

- ▶ **Key idea:** Subtract conditional means to remove the nuisance:

$$Y_i - m(X_i) = \tau(W_i - e(X_i)) + \epsilon_i$$

where  $m(x) = \mathbb{E}[Y|X = x]$ ,  $e(x) = \mathbb{E}[W|X = x]$ , and  $\mathbb{E}[\epsilon_i|X_i, W_i] = 0$ .

- ▶ **Estimation — residual-on-residual regression:**

1. Estimate  $\hat{m}(x)$  and  $\hat{e}(x)$  (e.g., with random forests)
2. Compute residuals  $\tilde{Y}_i = Y_i - \hat{m}(X_i)$ ,  $\tilde{W}_i = W_i - \hat{e}(X_i)$
3. OLS of  $\tilde{Y}_i$  on  $\tilde{W}_i$ :

$$\hat{\tau} = \frac{\sum \tilde{Y}_i \tilde{W}_i}{\sum \tilde{W}_i^2}$$

- ▶ **Properties:** Consistent if  $\hat{m}$  or  $\hat{e}$  is consistent (*double robustness*)

## Quiz 4A: Forest Interpretation

Random forests can be viewed as a data-adaptive kernel that assigns what to each training observation?

1. A binary leaf ID
2. Relevance weights  $\alpha_i(x)$
3. A predicted propensity score
4. A standardized residual

## Quiz 4A Solution

1. A binary leaf ID
2. Relevance weights  $\alpha_j(x)$
3. A predicted propensity score
4. A standardized residual

## Quiz 4B: Causal Forest Splitting

To target treatment heterogeneity, what does a Causal Forest maximize when selecting a split?

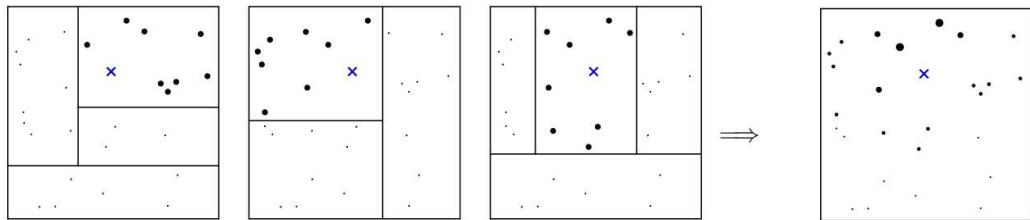
1.  $\sum (Y_i - \bar{Y})^2$
2.  $n_L n_R (\hat{\mu}_L - \hat{\mu}_R)^2$
3.  $n_L n_R (\hat{\tau}_L - \hat{\tau}_R)^2$
4.  $\frac{1}{n} \sum (W_i - \hat{e}(X_i))^2$

## Quiz 4B Solution

1.  $\sum (Y_i - \bar{Y})^2$  — standard regression tree
2.  $n_L n_R (\hat{\mu}_L - \hat{\mu}_R)^2$  — targets outcome, not treatment effect
3.  $n_L n_R (\hat{\tau}_L - \hat{\tau}_R)^2$  — maximizes difference in estimated treatment effects
4.  $\frac{1}{n} \sum (W_i - \hat{e}(X_i))^2$



## Forests as Adaptive Weights



Forests induce a data-adaptive weighting function. Training points sharing a leaf with  $x$  receive higher weights  $\alpha_i(x)$ .

## Forests as Adaptive Weights

- ▶ **Idea:** A random forest partitions feature space across many trees. For a test point  $x$ , each training unit  $i$  receives a weight:

$$\alpha_i(x) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}[x \text{ and } X_i \text{ in same leaf in tree } b]$$

Units that co-occur with  $x$  in more leaves get higher weight — the forest acts as a data-adaptive kernel.

- ▶ **Properties:**  $\alpha_i(x) \geq 0$ ,  $\sum_i \alpha_i(x) = 1$
- ▶ **Localized Robinson regression:**

$$\hat{\tau}(x) = \frac{\sum_i \alpha_i(x) (Y_i - \hat{m}(X_i)) (W_i - \hat{e}(X_i))}{\sum_i \alpha_i(x) (W_i - \hat{e}(X_i))^2}$$

This is Robinson's estimator applied locally using forest weights — confounding-robust by construction.

# Causal Forest: Splitting and Robustness

- ▶ **Splitting criterion:** Unlike standard regression trees (which maximize outcome variance), causal forests maximize heterogeneity in *treatment effects*:

$$\max_{j,s} n_L n_R (\hat{\tau}_L - \hat{\tau}_R)^2$$

Splits are chosen where  $\hat{\tau}$  differs most between left and right child nodes.

- ▶ **Two-stage procedure:**

1. Grow causal forest  $\Rightarrow$  obtain adaptive weights  $\alpha_i(x)$
2. Apply localized Robinson regression using estimated  $\hat{m}(x)$  and  $\hat{e}(x)$

- ▶ **Advantages over meta-learners:**

- ▶ Adapts to local confounding structure automatically
- ▶ Handles high-dimensional  $X$  without explicit modeling
- ▶ Provides uncertainty estimates via forest mechanics

# Key Formulas Summary

► **CATE:**  $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$

► **Meta-Learners:**

► S:  $\hat{\tau}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$

► T:  $\hat{\tau}(x) = \hat{\mu}^{(1)}(x) - \hat{\mu}^{(0)}(x)$

► X:  $\hat{\tau}(x) = \hat{e}(x)\hat{\tau}^{(1)}(x) + (1 - \hat{e}(x))\hat{\tau}^{(0)}(x),$

$$D_i^{(1)} = Y_i - \hat{\mu}^{(0)}(X_i), \quad D_i^{(0)} = \hat{\mu}^{(1)}(X_i) - Y_i$$

► **Causal Forest (localized Robinson):**

$$\hat{\tau}(x) = \frac{\sum_i \alpha_i(x)(Y_i - \hat{m}(X_i))(W_i - \hat{e}(X_i))}{\sum_i \alpha_i(x)(W_i - \hat{e}(X_i))^2}$$

# Key Takeaways

- ▶ **Regularization bias:** ML regularization affects treatment/control groups differently  $\Rightarrow$  can hallucinate heterogeneity.
- ▶ **Confounding bias:** Non-random treatment assignment makes naive comparisons unreliable.
- ▶ **Meta-learners** offer flexible CATE estimation with different bias trade-offs:
  - ▶ S-learner: simple but prone to regularization bias
  - ▶ T-learner: flexible but may hallucinate heterogeneity
  - ▶ X-learner: reduces bias via imputation; needs good propensity scores
- ▶ **Robinson's transformation:** Residual-on-residual regression removes nuisance functions; doubly robust.
- ▶ **Causal forests:** Extend Robinson's estimator locally using adaptive forest weights — confounding-robust and high-dimensional-friendly.